

A Simple, Low-Cost Audio Mixer

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When someone starts a small musical band, it is always difficult to make those big investments in expensive electronic equipment such as mixers, special piano effects, recording equipment, power amplifiers, loudspeakers, etc, until the group becomes successful. So the obvious solution is to use the home audio entertainment system

for practice. Some home audio power amplifiers have an external input and output power in tens of watts, which is more than enough to start the practice sessions.

The only thing that is really needed, apart from guitars and microphones, is a simple sound mixer. Although there is a wide range of relatively low-cost mixers to choose from, presented here is a simple and useful solution that can be built and

made operational quickly, with almost no investment.

Circuit and working

The circuit of the simple low-cost audio mixer is shown in Fig. 1. It is built around five low-cost BC550 transistors (T1 through T5) and some common components. The circuit has a total of seven inputs divided into two groups. The inputs CON1 through CON4 are for guitars or any source with a high signal level, such as a CD/DVD player, Walkman or noise generator. These inputs have a fixed input resistance of more than 700 kilo-ohms. Potmeters (VR1 through VR4) are provided for volume control.

Inputs CON5 through CON7 are for dynamic microphones. The input

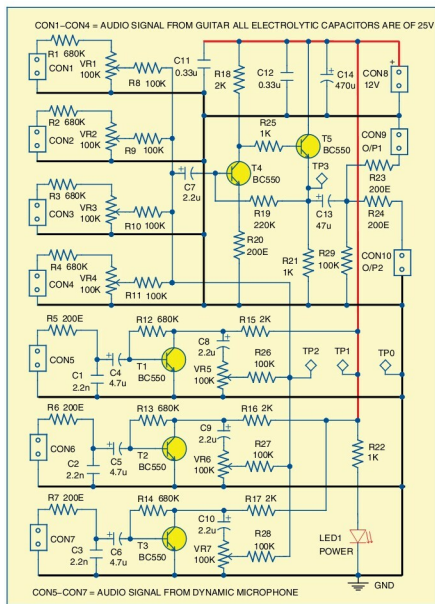


Fig. 1: Circuit of the simple low-cost audio mixer

Test Points

Test point	Details
TP0	0V, GND
TP1	+12V
TP2	Mixed audio (depending on the components connected on the input connectors)
TP3	Amplified output (depending on the value of R18 and R20)

PARTS LIST

Semiconductors:

- T1-T5 - BC550 npn transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, 5% carbon):

- R1-R4, R12-R14 - 680 kilo-ohm
- R5-R7, R20, R23, R24 - 200 ohm
- R8-R11, R25-R29 - 100 kilo-ohm
- R15-R18 - 2 kilo-ohm
- R19 - 220 kilo-ohm
- R21, R22 - 1 kilo-ohm
- VR1-VR7 - 100 kilo-ohm potentiometer

Capacitors:

- C1-C3 - 2.2nF ceramic disk
- C7-C10 - 2.2uF, 25V electrolytic
- C4-C6 - 4.7uF, 25V electrolytic
- C11-C12 - 0.33uF ceramic disk
- C13 - 47uF, 25V electrolytic
- C14 - 470uF, 25V electrolytic

Miscellaneous:

- CON1-CON10 - 2-pin connector